

Exercise 1

Consider an uncertain system of the form

$$x(k+1) = Ax(k) + w(k) \quad (1)$$

with $w(k) \in \mathcal{W}$. Let $\mathcal{X}$ be a nominal positive invariant set for $z(k+1) = Az(k)$ and let $\mathcal{E}$ be a robust positive invariant set for (1). Show that $\mathcal{X} \oplus \mathcal{E}$ is a robust invariant set for (1).

Solution:

For robust positive invariance of the set $\mathcal{X} \oplus \mathcal{E}$, we need to show that any state $x \in \mathcal{X} \oplus \mathcal{E}$, it holds that $Ax + w \in \mathcal{X} \oplus \mathcal{E}$ for all $w \in \mathcal{W}$.

- Split any $x(k) \in \mathcal{X} \oplus \mathcal{E}$ into $x(k) = z(k) + e(k)$ with $z(k) \in \mathcal{X}$ and $e(k) \in \mathcal{E}$ (see definition Minkowski sum).
- Insert into dynamics:

$$\begin{aligned} x(k+1) &= Ax(k) + w(k) \\ &= \underbrace{Az(k)}_{z(k+1)} + \underbrace{Ae(k) + w(k)}_{e(k+1)} \end{aligned}$$

- We have $z(k) \in \mathcal{X} \Rightarrow z(k+1) \in \mathcal{X}$ (nominal invariance of $\mathcal{X}$) and $e(k) \in \mathcal{E} \Rightarrow e(k+1) \in \mathcal{E}$ (robust invariance of $\mathcal{E}$), which implies that $x(k+1) = z(k+1) + e(k+1) \in \mathcal{X} \oplus \mathcal{E}$.

Exercise 2

Let $X = [a, b]$, $Y = [c, d]$ with finite $a, b, c, d \in \mathbb{R}$, $a < 0 < b$, $c < 0 < d$, and $Y \subset X$. In this case, the Minkowski sum $X \oplus Y$ and the Pontryagin set difference $X \ominus Y$ yield both again compact sets, and $X \oplus Y$ and $X \ominus Y$ both yield sets of the form $[e, f]$, $e \leq 0 \leq f$. Show that

- i) $X \oplus Y = [a + c, b + d]$
- ii) $X \ominus Y = [a - c, b - d]$

Solution:

- i) Let $x, y \in \mathbb{R}$. By definition

$$\begin{aligned} X \oplus Y &= \{x + y \mid x \in \mathcal{X}, y \in \mathcal{Y}\} \\ &= \{x + y \mid x \in [a, b] \wedge y \in [c, d]\}. \end{aligned}$$

Since $e \leq f$ and $X \oplus Y = [e, f]$ we calculate

$$\begin{aligned} e &= \min_{x \in \mathcal{X}, y \in \mathcal{Y}} x + y \\ &= a + c \end{aligned}$$

and

$$\begin{aligned} f &= \max_{x \in \mathcal{X}, y \in \mathcal{Y}} x + y \\ &= b + d. \end{aligned}$$

ii) By definition

$$\begin{aligned} X \ominus Y &= \{z \in \mathbb{R} \mid z + y \in X, \forall y \in Y\} \\ &= \{z \in \mathbb{R} \mid z + y \in [a, b] \forall y \in [c, d]\}. \end{aligned}$$

Since $e \leq f$ and $X \ominus Y = [e, f]$ we solve

$$\begin{aligned} e &= \min_{z \in \mathbb{R}} z \\ \text{s.t. } z + y^* &\in X \\ y^* &= \min_{y \in Y} y \end{aligned}$$

and

$$\begin{aligned} f &= \max_{z \in \mathbb{R}} z \\ \text{s.t. } z + y^* &\in X \\ y^* &= \max_{y \in Y} y \end{aligned}$$

by calculating the extreme points

$$\begin{aligned} c &= \min_{y \in Y} y \\ d &= \max_{y \in Y} y \end{aligned}$$

we get the lower bound

$$\begin{aligned} e &= \min_{z+c \in X} z \\ &= a - c. \end{aligned}$$

as well as the upper bound

$$\begin{aligned} f &= \max_{z+d \in X} z \\ &= b - d. \end{aligned}$$

Exercise 3

Let $\mathcal{A}_1 := \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 2\}$, $\mathcal{A}_2 := \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$. and $\mathcal{B} := \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}$. Graphically compute

i) $\mathcal{A}_1 \ominus \mathcal{B} \oplus \mathcal{B}$

ii) $\mathcal{A}_2 \ominus \mathcal{B} \oplus \mathcal{B}$

What do you observe?

Solution:

i) $\mathcal{A}_1 \ominus \mathcal{B} \oplus \mathcal{B} = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\} \oplus \mathcal{B}$

ii) $\mathcal{A}_2 \ominus \mathcal{B} \oplus \mathcal{B} = \mathcal{B}$